

strong nuclear interaction

Mass defect $\implies$ the mass of a nucleus $<$ total mass of its constituent parts

$\implies$ nuclear binding energy

$\therefore$ energy and mass are considered interchangeable:

$$E = mc^2$$

where c is the speed of light

Binding energy originates inside the nucleus. **Protons are positively charged** and repel each other through [Coulomb's law](#).

Hence, there must be an additional attractive interaction that opposes this repulsion to **hold the nucleus together**.

An attractive **strong nuclear interaction** (strong nuclear force) acts between the protons and neutrons.

At **short distances**,

strong nuclear force is **repulsive**;

protons and neutrons are repelled

At **long distances**,

strong nuclear force is **attractive**;

protons and neutrons are attracted to each other;

it binds the protons and neutrons together

At distance of $1 \text{ fm} = 1 \times 10^{-15} \text{ m}$

strong attraction $\approx 100 \times$ EM repulsion

Energy must be added to a nucleus to separate it into its component parts.

Hence, to form a nucleus by bringing its individual nucleons together from infinity, energy must be **transferred out of the nucleus**. This links to [electric potential](#).

This is the source of **binding energy = mass defect**.